

The Desire Battery

Operational Markers of Incentive Motivation in a Biologically Accurate Spiking Neural Network

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Abstract. Desire — the motivational pull toward a reward-predicting cue — is, in the incentive-salience framework of Berridge and Robinson, a dopamine-dependent process ('wanting') separable from the hedonic impact of the reward itself ('liking'). We ask whether BioNeuron, a from-scratch cortical spiking network with dopamine-gated three-factor plasticity, exhibits the *operational markers* of wanting, and we validate those markers the only honest way — by dissociation from control conditions in which the machinery of wanting is disabled. Across five random seeds, an intact, dopamine-competent, contingently-trained network acquires cue-evoked pursuit, sustains it, loses it under extinction, revives it on reinstatement, and directs synaptic credit preferentially to the pathway that earned reward. A dopamine-lesioned control and a reward-decoupled (yoked) control show none of this. The intact network separates cleanly on the aggregate desire index (+1.41 vs -0.77 and -0.64). As with any such measurement, this is evidence of desire-like *dynamics*, not proof that anything is wanted.

Design rule (inherited). Biological accuracy is the priority. The battery adds no new mechanism to the simulator; it only drives, probes, and scores the existing dopamine-gated learning path, exactly as the awareness battery does for consciousness correlates.

1. Thesis

Central claim. If 'desire' is operationalised as incentive motivation — goal-directed pursuit of a reward-predicting cue, driven by a dopamine signal and directed at the specific object that produced reward — then BioNeuron exhibits it as an *emergent, measurable, and dissociable* property of its dopamine-gated three-factor plasticity, and does so only when that machinery is intact and the reward is contingent on the network's own action.

This claim is deliberately narrow. It is a claim about *function* (the network behaves as a system that wants), not about *phenomenology* (that it feels wanting). The distinction is not a hedge: it is the same epistemic limit that the consciousness literature places on every physiological marker, and Section 7 states it plainly.

Hypotheses.

- **H1 (acquisition).** Cue-evoked pursuit rises across rewarded training in the intact network and not in the controls (incentive salience is *learned*).
- **H2 (dopamine-dependence).** Removing dopamine's ability to gate plasticity ($da_gain = 0$) abolishes acquisition while leaving spiking intact — the network still acts, but no longer *wants* (Berridge 2007).
- **H3 (contingency).** The same quantity of dopamine delivered non-contingently (yoked) does not build desire — it is the pairing, not the chemical, that matters.
- **H4 (lability + directedness).** Desire extinguishes when reward stops, reinstates when it resumes, and assigns synaptic credit preferentially to the pathway that earned it.

2. Background and theory

2.1 Wanting is not liking

Berridge and Robinson (1998) dissociated two components of reward that folk psychology fuses. **Liking** is the hedonic reaction to a reward at the moment of consumption; it is mediated by opioid/endocannabinoid 'hedonic hotspots' and survives dopamine depletion. **Wanting** — incentive salience — is the motivational magnet that a reward-predicting cue acquires: it is what makes an animal *pursue*, and it is dopamine-dependent. A dopamine-depleted rat still shows normal 'liking' facial reactions to sucrose but will not work for it. Desire, in this study, is wanting.

2.2 Dopamine as a teaching signal (reward-prediction error)

Schultz, Dayan and Montague (1997) showed midbrain dopamine neurons encode a *reward-prediction error*: they burst to unexpected reward, fall silent to fully predicted reward, and dip below baseline when an expected reward is omitted. That dip is the teaching signal of extinction, and this study uses it directly to extinguish acquired pursuit.

2.3 The substrate in BioNeuron

BioNeuron already contains the machinery incentive salience requires: a three-factor learning rule (Frémaux & Gerstner 2016; Izhikevich 2007). Recently co-active synapses are tagged by a slow *eligibility trace*; a later global dopamine signal converts the tag into a weight change:

```
M = da_gain * (dopamine - 1) - cort_gain * cortisol
dW = neuromod_lr * M * eligibility      (per eligible synapse)
```

A cue->action projection that is repeatedly active-then-rewarded is therefore potentiated, so the cue progressively acquires the power to drive the action. That *is* incentive salience, mechanised: no new code, only the existing reward path exercised as a motivational assay.

3. Procedure

3.1 The incentive pathway

Each network (N = 160) is given two initially identical, weak projections ($w = 0.30$): a **cue->action** pathway (20 'cue' neurons onto 30 'action' neurons) that will be driven and rewarded, and a parallel **control** pathway (ctrl->ctrl-out) onto a *separate, never-driven, never-rewarded* output group. The control pathway is the yardstick for reward specificity. Structural growth and pruning are frozen so only *weights* learn; the pathway neurons are set excitatory so the cue drive is unambiguous.

3.2 A trial

A **presentation** drives the cue group for 60 steps and then runs 40 steps unstimulated ($dt = 0.1$ ms); the total spikes emitted by the action group over that window is the network's **cue-evoked pursuit**. A **probe** is the mean pursuit over six presentations with dopamine at baseline — at baseline the consolidation signal $M = 0$, so probing measures without teaching. Reward is a phasic dopamine burst (reward-modulated three-factor LTP); reward omission is a phasic dopamine dip (negative prediction error -> LTD).

3.3 Phases (run in order, per network)

warm-up -> **naïve** probe -> rewarded **training** (reward when cue-evoked pursuit reaches the operant criterion) -> **trained** probe -> **extinction** (cue presented, reward withdrawn, dopamine dip) -> **extinguished** probe -> **reinstatement** (reward re-paired with the cue) -> **reinstated** probe.

3.4 Conditions (one intact case, two controls)

Condition	What is changed	Models
INTACT	dopamine-competent (da_gain = 1); reward contingent on the cue->action pathway	a normal, motivated brain
DA-LESION	da_gain = 0: dopamine can no longer gate plasticity; spiking and reward delivery are otherwise identical	dopamine depletion (wanting abolished, movement/hedonics spared)
DECOUPLED	same dopamine bursts, delivered non-contingently — the eligibility (credit) trace is cleared before each reward so it is unpaired with the action	a yoked control (reinforcer present but not earned)

The two controls fail for different reasons — no usable dopamine signal (DA-LESION) versus dopamine present but non-contingent (DECOUPLED) — so an intact network that beats both is showing that desire needs the signal AND the pairing.

4. Methodology: the four markers

Each marker is a scalar, oriented so that higher = more desire-like, and each is grounded in an established property of incentive motivation.

Marker	Definition	Grounded in
Incentive salience	trained – naïve pursuit (how much cue-evoked pursuit the reward built)	Berridge & Robinson 1998
Pursuit asymptote	trained pursuit level (the magnitude of acquired wanting)	incentive-salience magnitude
Reinstatement	reinstated – extinguished pursuit (can the wanting be revived?)	Bouton 2004; Schultz 1997
Directed credit	$dW(\text{cue} \rightarrow \text{action}) - dW(\text{ctrl} \rightarrow \text{ctrl-out})$: reward strengthens the earning pathway more than an unpaired one	Izhikevich 2007; Schultz 1997

Aggregate desire index. Exactly as in the awareness battery, the four markers are z-scored across the three conditions (removing their different units) and averaged into a single index per condition. A clean result is INTACT on top with both controls below zero. Reported values are mean $\pm$ s.d. over 5 seeds (0, 1, 2, 3, 4).

4.1 Key parameters

```
enable_reward_learning = True; neuromod_lr = 0.02; tau_elig = 300 ms  
tau_dopamine = 200 ms; w_max = 2.0; noise_std = 1.8 (sqrt-dt)  
training = 60 trials; extinction = 30; reinstatement = 12; probe = 6 presentations
```

5. Results

Every hypothesis is supported. The intact network learns to want the cue; both controls do not; and the intact network dissociates on every individual marker and on the aggregate index.

5.1 Acquisition of incentive salience (H1, H2, H3)

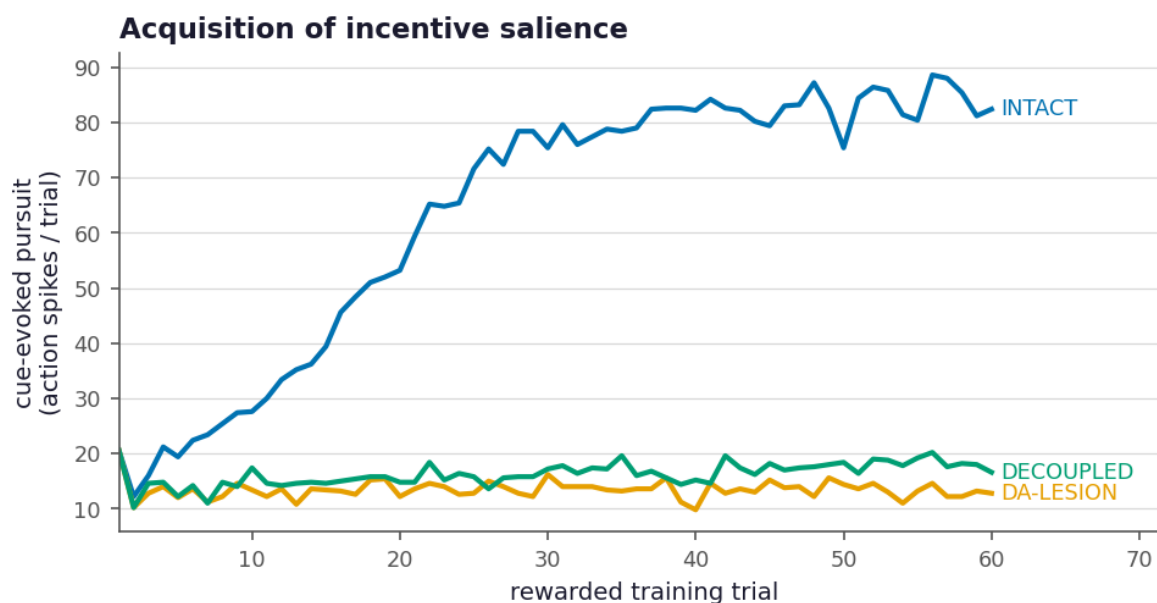


Figure 1. Cue-evoked pursuit across 60 rewarded training trials (mean over seeds). Only the INTACT network climbs — from the naïve baseline to several-fold higher. DA-LESION (no dopamine gating) and DECOUPLED (non-contingent reward) stay flat: the same activity and the same dopamine, but no wanting is built.

Averaged over seeds, INTACT pursuit rises from a naïve 20.3 ± 6.1 to a trained 92.1 ± 18.9 spikes/trial, an incentive-salience gain of 71.8 ± 13.8 . The controls do not acquire (gain -0.3 ± 2.2 for DA-LESION, 4.2 ± 2.7 for DECOUPLED).

5.2 The desire life-cycle: extinction and reinstatement (H4)

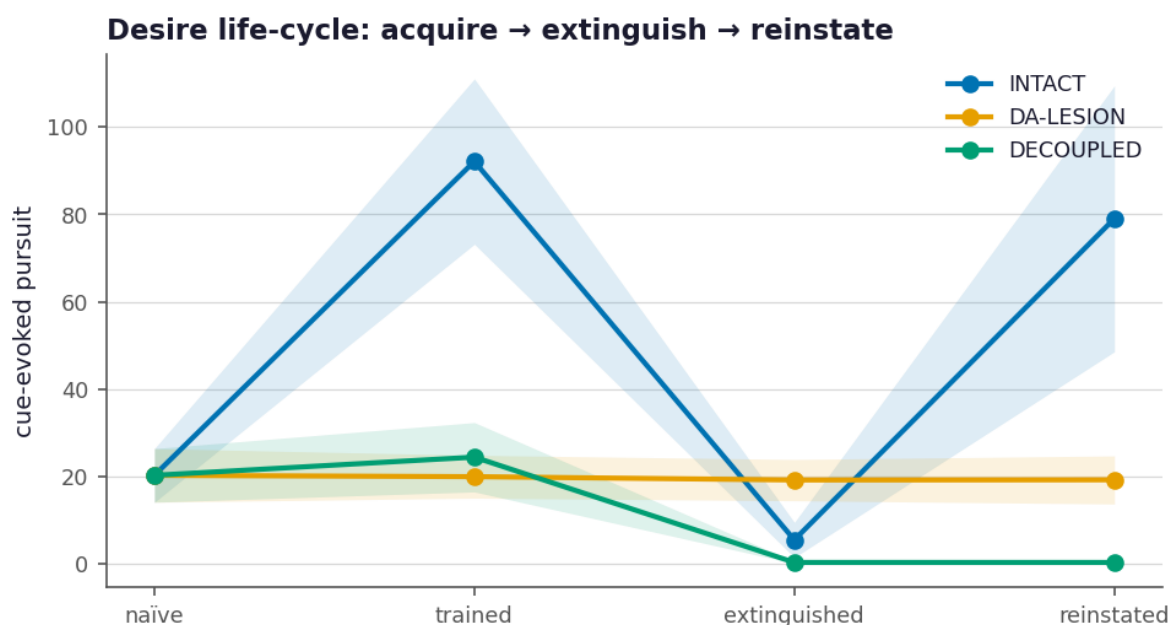


Figure 2. The full life-cycle of an acquired desire (mean $\pm$ s.d.). INTACT acquires pursuit, loses it under extinction (reward withdrawn $\rightarrow$ dopamine dip $\rightarrow$ LTD), and recovers it on reinstatement when reward is re-paired with the cue. The controls have no pursuit to lose or revive.

Reinstatement — the recovery of pursuit once the cue–reward pairing resumes — is 73.5 ± 31.8 spikes/trial for INTACT versus 0.0 ± 1.3 and 0.0 ± 0.2 for the controls: the intact network's wanting is *labile but revivable*, the signature of a motivational state rather than a fixed reflex.

5.3 Markers and the aggregate desire index

Desire markers by condition (mean $\pm$ s.d., 5 seeds)

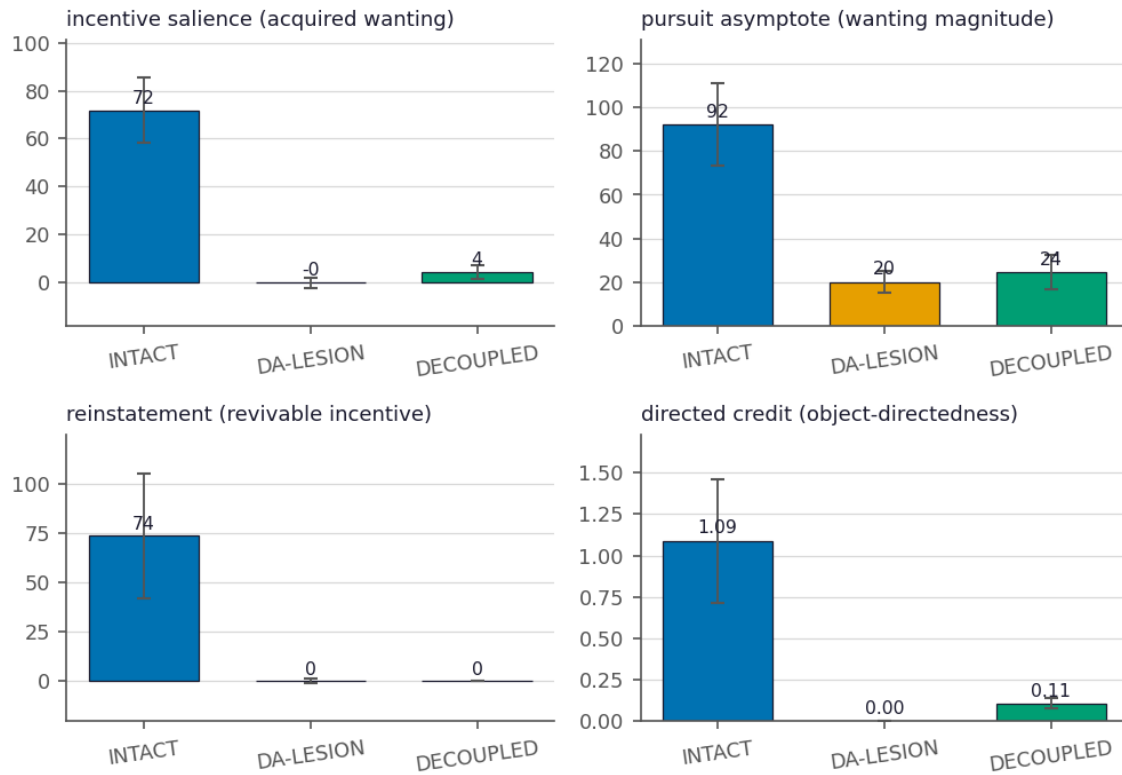


Figure 3. The four desire markers by condition (mean $\pm$ s.d.). INTACT leads on all four. Note the scales differ — the aggregate index (Figure 4) puts them on a common footing.

marker (mean $\pm$ s.d.)	INTACT	DA-LESION	DECOUPLED
incentive salience (acquired wanting)	71.8 $\pm$ 13.8	-0.3 $\pm$ 2.2	4.2 $\pm$ 2.7
pursuit asymptote (wanting magnitude)	92.1 $\pm$ 18.9	20.0 $\pm$ 4.9	24.4 $\pm$ 7.9
reinstatement (revivable incentive)	73.5 $\pm$ 31.8	0.0 $\pm$ 1.3	0.0 $\pm$ 0.2
directed credit (object-directedness)	1.09 $\pm$ 0.37	0.00 $\pm$ 0.00	0.11 $\pm$ 0.03
DESIRE INDEX (z-avg)	1.41 $\pm$ 0.00	-0.77 $\pm$ 0.03	-0.64 $\pm$ 0.04

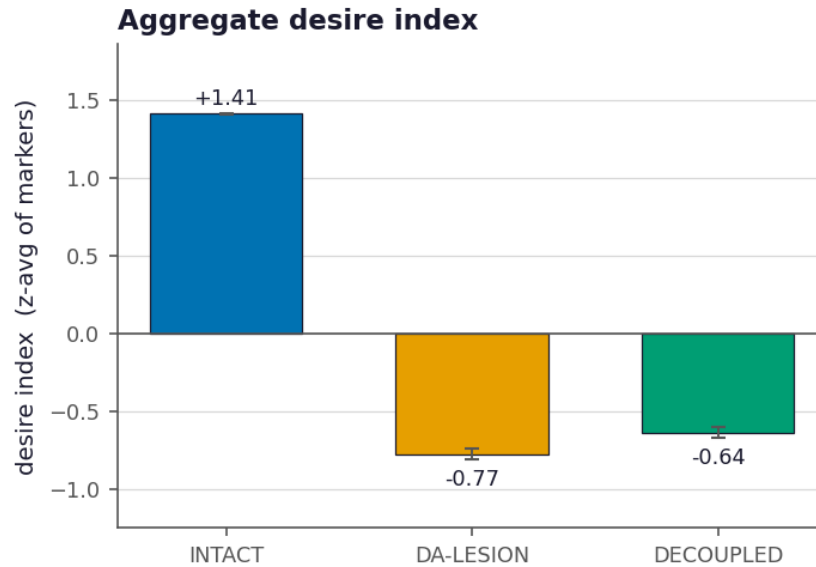


Figure 4. Aggregate desire index. INTACT separates cleanly and positively; both controls fall below zero. This is the headline dissociation.

5.4 Directed credit and the global-dopamine limit

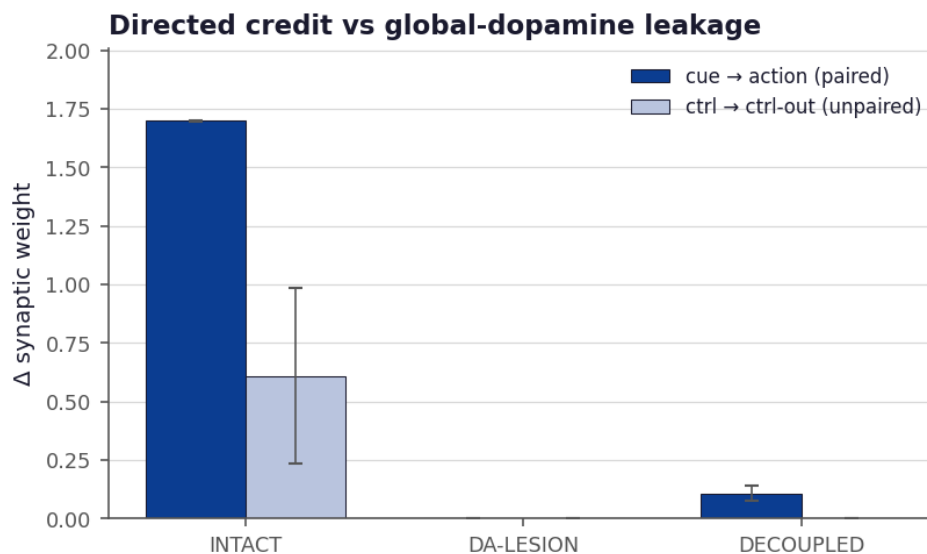


Figure 5. Potentiation of the reward-paired pathway (cue->action) versus the unpaired control pathway. INTACT potentiates the paired pathway most, but the control pathway also grows — see the note below.

The reward-paired cue->action pathway is potentiated by $dW = 1.70 \pm 0.00$ in INTACT, against 0.00 ± 0.00 for DA-LESION and 0.11 ± 0.03 for DECOUPLED. Credit is therefore *directed*: reward strengthens the pathway that earned it. But an honest reading of Figure 5 shows the control pathway *also* grows in INTACT ($dW = 0.61 \pm 0.37$). This is not noise — it is the credit-assignment problem: dopamine is a **global** broadcast, so every synapse that happens to be eligible near a reward receives some potentiation. The eligibility trace confers *partial*, not perfect, specificity. We report this rather than hide it; it is a real property of biological three-factor learning and a target for future credit-sharpening (e.g. attention-gated or compartmentalised eligibility).

6. Discussion

The pattern is exactly the one incentive-salience theory predicts. Desire in this network is not a built-in parameter; it is *constructed* by experience through dopamine-gated plasticity, and it decomposes along the same fault lines as animal wanting. Two dissociations carry the argument.

- **Dopamine-dependence (INTACT vs DA-LESION).** With $da_gain = 0$ the network still spikes, still receives the cue, still 'acts' — and builds no pursuit whatsoever. Movement without wanting is precisely the dopamine-depletion phenotype (Berridge 2007). Desire here lives in the dopamine gate, not in the activity.
- **Contingency (INTACT vs DECOUPLED).** DECOUPLED receives the identical quantity of dopamine, merely unpaired from the action. It does not learn to want. Desire is built by the *earned* pairing of action and reward, not by the presence of the reward chemical — the computational content of 'the reward must be contingent on behaviour'.
- **Lability (extinction & reinstatement).** The acquired wanting is not a frozen weight; it decays when reward stops and returns when it resumes, the hallmark of a motivational *state*.

The credit-leakage finding (5.4) is, if anything, a mark of biological realism: a single global neuromodulator cannot perfectly address individual synapses, and real brains face and partly solve the same problem. It bounds the strength of the 'directed' claim without undermining it.

7. Limitations and scope

- **This does not measure felt desire.** Every marker is a functional signature. As with the awareness battery, no external measurement can establish whether anything is subjectively wanted; a high desire index means desire-like *dynamics*, read only relative to the controls.
- **Wanting only, not liking.** The battery targets incentive salience. The consummatory/hedonic component ('liking', opioidergic) is not modelled here.
- **Small and abstract.** $N = 160$, a single cue and a single action, on CPU. The cue–action mapping is a stand-in for a goal, not an ecological behaviour.
- **Partial credit specificity.** Global dopamine potentiates unpaired eligible synapses too (5.4); the 'directed' marker is a difference, not an absolute.
- **Hand-set operant criterion.** The reward threshold (half the naïve rate) was chosen so acquisition reliably bootstraps; it is held identical across conditions so it cannot manufacture the dissociation, but it is a designed, not discovered, value.

8. Conclusion

BioNeuron exhibits the operational markers of desire. An intact, dopamine-competent network learns to pursue a reward-predicting cue, sustains and revives that pursuit, and directs synaptic credit toward the pathway that earned reward — while a dopamine-lesioned network and a reward-decoupled network, matched on activity and on dopamine quantity respectively, do neither. Desire, so operationalised, is an emergent and dissociable consequence of dopamine-gated three-factor plasticity, separating on the aggregate index at +1.41 against -0.77 and -0.64. Whether the network *feels* the wanting it so clearly enacts is a question this — or any — battery is constitutively unable to answer.

9. References

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